Control work 3 · Roots and rational exponents

The Chapter I assessment, followed by a work-on-mistakes lesson sorted by error type.

LESSON 36–37 | 2 × 40 MIN | ALGEBRA 8, §§8–10 | STAGE 9 · UNIT 1 CHECK

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess roots, rational exponents, standard form and decimals.
- Sort the errors by type rather than by question.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of §§8–10
Cambridge	Learner's Book pp. 9–20
Workbook	Workbook Unit 1

01 Explanation

Lesson 36 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each, 14 in total:

- tasks 1–2 · simplify a root (§8)
- task 3 · rationalise a denominator (§8)
- tasks 4–5 · evaluate a power with a rational exponent (§9)
- task 6 · simplify a letter expression (§10)
- task 7 · standard form or a recurring decimal (Cambridge insert)

Lesson 37 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. **Splitting a root over a sum** — writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.
2. **Numerator and denominator of the index swapped** — reading $8^{2/3}$ as $(8^3)^2$.
3. **A negative index treated as a negative answer** — $4^{-1/2}$ given as -2.
4. **Standard form not tidied** — leaving 12×10^9.

The Hard set below is seven pieces of wrong working built from these four.

02 Worked examples

EXAMPLE 1 **Find the error:** $\sqrt{16 + 9} = 4 + 3 = 7$

1. A root does not split over a sum.
   Only over a product or a quotient.

2. $16 + 9 = 25$
   Do the addition first.

3. $\sqrt{25} = 5$

ANSWER

5

EXAMPLE 2 **Find the error:** $4^{-1/2} = -2$

1. A negative index means a reciprocal, not a negative number.

2. $4^{-1/2} = \dfrac{1}{4^{1/2}}$

3. $= \dfrac{1}{2}$

ANSWER

$\dfrac{1}{2}$

03 Interactive model

Show each piece of wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: $\sqrt{16 + 9} = 7$

1. The product rule is $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$. There is no sum rule.

2. $16 + 9 = 25$, so the root is $\sqrt{25}$.

3. $= 5$, not 7. One counter-example is enough to kill the false rule.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Root	Ildiz	Корень
Exponent	Ko‘rsatkich	Показатель
Standard form	Standart ko‘rinish	Стандартный вид
Evaluate	Hisoblash	Вычислить
Justify	Asoslash	Обосновать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sqrt{25 + 144}$ equals:

- 17
- 13
- 169
- $5 + 12$

$8^{2/3}$ equals:

- 4
- 64
- 16
- 262144

$9^{-1/2}$ equals:

- -3
- $\frac{1}{3}$
- 3
- $-\frac{1}{3}$

20×10^9 in standard form is:

- 20×10^9
- 2×10^{10}
- 2×10^8
- 0.2×10^{11}

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. ***Variant 1, task 1.*** *Simplify* $\sqrt{48}$

   ANSWER $4\sqrt{3}$

2. ***Task 2.*** *Simplify* $\sqrt{6} \cdot \sqrt{24}$

   ANSWER 12

3. ***Task 3.*** *Rationalise* $\dfrac{4}{\sqrt{2}}$

   ANSWER $2\sqrt{2}$

4. ***Task 4.*** *Evaluate* $16^{1/2}$

   ANSWER 4

5. ***Task 5.*** *Evaluate* $27^{2/3}$

   ANSWER 9

6. ***Task 6.*** *Simplify* $a^{1/2} \cdot a^{3/2}$

   ANSWER a^2

7. ***Task 7.*** *Write 0.0035 in standard form*

   ANSWER 3.5×10^{-3}

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Simplify* $\sqrt{75}$

   ANSWER $5\sqrt{3}$

2. ***Task 2.*** *Simplify* $\sqrt{8} \cdot \sqrt{18}$

   ANSWER 12

3. ***Task 3.*** *Rationalise* $\dfrac{9}{\sqrt{3}}$

   ANSWER $3\sqrt{3}$

4. ***Task 4.*** *Evaluate* $64^{1/3}$

   ANSWER 4

5. ***Task 5.*** *Evaluate* $25^{-1/2}$

   ANSWER $\dfrac{1}{5}$

6. ***Task 6.*** *Simplify* $\dfrac{b^{5/4}}{b^{1/4}}$

   ANSWER b

7. ***Task 7.*** *Write* $\dfrac{2}{3}$ *as a decimal*

   ANSWER $0.(6)$

Hard · stretch and reason

7 PROBLEMS

1. *Find the error:* $\sqrt{16+9} = 7$

ANSWER Roots do not split over a sum. Correct: 5.

2. *Find the error:* $8^{2/3} = (8^3)^2$

ANSWER Index upside down. Correct: 4.

3. *Find the error:* $4^{-1/2} = -2$

ANSWER A negative index gives a reciprocal. Correct: $\frac{1}{2}$.

4. *Find the error:* $(4 \times 10^5)(5 \times 10^4) = 20 \times 10^9$

ANSWER Not tidied. Correct: 2×10^{10}.

5. *Find the error:* $\sqrt{a^2} = a$ *for every* a

ANSWER True only for $a \geq 0$; for $a < 0$ it is $-a$.

6. *Find the error:* $\frac{1}{\sqrt{2}} = \sqrt{2}$

ANSWER Rationalising gives $\frac{\sqrt{2}}{2}$, which is about 0.71, not 1.41.

7. *Find the error:* $0.(3) = \frac{3}{10}$

ANSWER $\frac{3}{10} = 0.3$ exactly, not recurring. Correct: $\frac{1}{3}$.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on, then these three.

1. *Simplify* $\sqrt{147}$ *and rationalise* $\frac{5}{\sqrt{5}}$
2. *Evaluate* $32^{3/5}$ *and* $49^{-1/2}$

3. Write out the four errors from this lesson in your own words with one example each.